90
Pass

Deep Panchal

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Score

90% • 157.5 / 175
scored in eClerx - Java Coding Test - 3+ Years New in 50 min 16 sec on 18 Feb 2026 11:53:16 IST

Candidate Information

Email	panchaldeep360@gmail.com
Test	eClerx - Java Coding Test - 3+ Years New
Candidate Packet	View
Taken on	18 Feb 2026 11:53:16 IST
Time taken	50 min 16 sec/ 90 min
Work Experience	3 years
Invited by	Vipul
Invited on	18 Feb 2026 09:02:34 IST

Skill Distribution

No.	Skill	Score
1	Spring Boot Advanced	0%

2	Spring Boot Intermediate	75%
3	Problem Solving Intermediate	100%
4	Java Intermediate	100%

Tags Distribution

Spring Boot	30%	Hard	0%
TDD	0%	Spring	50%
JPA	50%	Spring Data	50%
Transactional	50%	Medium	98%
Implementation	100%	Arrays	100%
Algorithms	100%	Problem Solving	100%
Greedy	100%	Java	100%

Questions

Spring Boot (Advanced) • 7.5 / 25

No.	Question	Skill	Score	Code Quality	Time Taken	Status

1	JSON Deserialization Multiple Choice	Spring Boot (Advanced)	0/5	-	1 min 42 sec	⊗
2	Determine the Output Multiple Choice	Spring Boot (Advanced)	0/5	-	5 min 12 sec	⊗
3	Output Multiple Choice	Spring Boot (Advanced)	0/5	-	1 min 28 sec	⊗
4	Spring Data Transactional Multiple Choice	Spring Boot (Intermediate)	2.5/5	-	3 min 56 sec	✓
5	Endpoint Security Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 9 sec	✓

Problem Solving (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
6	Wind Farms Coding	Problem Solving (Intermediate)	75/75	-	18 min 11 sec	✓

Java (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status

7	Java: Optimized Sequential Video Watching Coding	Java (Intermediate)	75/75	-	17 min 44 sec	✓

Candidate Feedback

Nice UI

1. JSON Deserialization

Incorrect

Multiple Choice | Spring Boot | Hard

Question description

CheckerCreateRequest.java

```
// Annotation here
public class CheckerCreateRequest {
    private String name;
    private String surname;
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getSurname() {
        return surname;
    }
    public void setSurname(String surname) {
        this.surname = surname;
    }
}
```

CheckerController.java

```
@RestController
@RequestMapping("/checkers")
```

```
public class CheckerController {

    @Autowired
    CheckerService checkerService;

    @PostMapping
    public ResponseEntity createChecker(@RequestBody CheckerCreateRequest checkerCreateReque
st) {
        checkerService.createChecker(checkerCreateRequest);
        return new ResponseEntity(HttpStatus.CREATED);
    }
}
```

Request Json data

```
{
    "checker":{
        "name": "Checker",
        "surname":"Number2"
    }
}
```

A POST request with "Request Json data" is made to "/checkers". What annotation needs to be added to the "CheckerCreateRequest" class so that it can successfully deserialize and process the request?

Interviewer guidelines

@JsonRootName doesn't add checker as a wrapper attribute by itself. To make it successfully deserialize with given json annotations on D should be used together.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ● @JsonRootName("checker")

- @JsonName("checker")
- @JsonTypeName("checker")
- @JsonTypeInfo(include = JsonTypeInfo.As.WRAPPER_OBJECT, use = JsonTypeInfo.Id.NAME)
  @JsonTypeName("checker")

No comments.

2. Determine the Output

Incorrect

Multiple Choice | Spring Boot | Hard

Question description

SpringQuestionApplication.java

```
package com.hackerrank.spring;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringQuestionsApplication {
   public static void main(String[] args) {
      SpringApplication.run(SpringQuestionsApplication.class, args);
   }
}
```

Lie.java

```
package com.hackerrank;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;
```

```
@Component
public class Lie implements CommandLineRunner {
   @Override
   public void run(String... args) throws Exception {
      System.out.println("Truth is behind the lies");
      System.out.println("This is working perfectly");
      throw new Exception("Does this work?");
   }
}
```

Truth.java

```
package com.hackerrank.spring.reality;

import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

@Component
public class Truth implements CommandLineRunner {
   @Override
   public void run(String... args) throws Exception {
      System.out.println("Truth is in your eyes");
   }
}
```

What will be printed on the console when the "SpringBootApplication" class is executed?

Interviewer guidelines

Correct Answer:

Option [A] – "Truth is in your eyes"

Explanation: Spring Boot scans only the package of the `@SpringBootApplication` class and its subpackages. `Truth` is under a subpackage and runs. `Lie` is in a parent package, so it is **not scanned** and its runner does not execute.

Incorrect Answers:

Option [B] – Multiple prints + exceptions

Explanation: `Lie` never runs, so its prints and exception do not occur.

Option [C] – Prints from Lie only

Explanation: `Truth` is discovered and executed; `Lie` is not.

Option [D] – Prints from Lie, then Truth
Explanation: `Lie` is outside the scan path and never executed.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ Truth is in your eyes ✓
- ○ Truth is in your eyes Truth is behind the lies This is working perfectly And exception occurs and application stops
- ○ Truth is behind the lies This is working perfectly And exception occurs and application stops
- ● Truth is behind the lies This is working perfectly And exception occurs containing “Does this work?” Truth is in your eyes

No comments.

3. Output

Incorrect

Multiple Choice | Hard | Spring Boot | TDD

Question description

What will happen to the output file?

```
@SpringBootTest
public class LogTest
{
   File log;
   FileWriter logWriter;

   @Autowired
   SomeClassToTest someClassToTest;

   @Before
   public void init()
   {
     log = new File("log.txt");
     logWriter = new FileWriter(log);
   }

   @After
   public void teardown()
   {
      log.delete();
   }

   @Test
   public void test1()
   {
      logWriter.write("Entered Test 1");
      assertTrue(someClassToTest.someMethod());
   }

   @Test
   public void test2()
   {
      logWriter.write("Entered Test 2");
      assertTrue(someClassToTest.otherMethod());
   }
}
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- The output file is not created.
- **The output file is created but not deleted.**
- The output file is created and eventually deleted.
- The create and delete blocks will not be executed.

No comments.

4. Spring Data Transactional

Partially correct

Multiple Choice | Spring | Spring Boot | JPA | Spring Data | Transactional | Medium

Question description

Given the following code, what happens when *StudentRepsitoryTest* runs?

Lesson.java

```
@Entity
public class Lesson {
    @Id
    private Long id;
    private String name;
    private Integer courseLength;
    private Integer courseLevel;
    @JsonBackReference
    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "student_id")
    private Student student;
```

```
    private boolean inProgress;
    //getter & setters
}
```

Student.java

```
@Entity
public class Student {
    @Id
    private Long id;
    private String name;
    private String surname;
    @JsonManagedReference
    @OneToMany(mappedBy = "student", fetch = FetchType.EAGER, cascade = {CascadeType.PERSIST,
CascadeType.REFRESH})
    private List<Lesson> lessons;
    //getter & setters
}
```

StudentRepository.java

```
@Repository
public interface StudentRepository extends JpaRepository<Student, Long>{
    //Y
    Optional<Student> findById(Long id);
}
```

LessonRepository.java

```
@Repository
public interface LessonRepository extends JpaRepository<Lesson, Long> {
}
```

StudentService.java

```
@Service
public class StudentService{
    private StudentRepository studentRepository;
    public StudentService(StudentRepository studentRepository, LessonRepository lessonRepository)
{
        this.studentRepository = studentRepository;
        this.lessonRepository = lessonRepository;
    }

    public void save(Student student) {
        studentRepository.save(student);
    }

    //other crud metods

    //X
    public void deleteStudentWithLessons(Long id) throws SQLException {
        Student student =
                studentRepository.findById(id).orElseThrow(() -> new RuntimeException("Student not foun
d"));
        List<Lesson> lessons = student.getLessons();
        lessons.stream().forEach(lesson -> lesson.setStudent(null));
        studentRepository.delete(student);

        boolean inProgress =lessons.stream()
                .map(Lesson::isInProgress)
                .filter(progress -> progress == true)
                .findFirst()
                .orElseGet(() -> false);
        if (inProgress) {
            throw new SQLException("Cannot delete all Lessons. There are lessons in progress state");
        }
        lessonRepository.deleteAll(lessons);
    }
}
```

StudentRepositoryTest.java

```
@SpringBootTest
class StudentRepositoryTest {
    @Autowired
    StudentRepository studentRepository;
    @Autowired
```

```
    StudentService studentService;
    @Autowired
    LessonRepository lessonRepository;
    Student student;
    @BeforeEach
    public void init() {
        student = Student.builder().name("Jorn").surname("Deynhoven").build();
        studentRepository.save(student);
        lessonRepository.saveAll(List.of(
                Lesson.builder().name("Music").courseLength(10).student(student).courseLevel(123).build
(),
                Lesson.builder().name("Art").courseLength(20).courseLevel(123).student(student).build(),
                Lesson.builder().name("Trance").courseLength(30).courseLevel(123).student(student).inPro
gress(true).build()
        ));
    }
    @Test
    void deleteStudentWithLessonsTest() throws SQLException {
        studentService.deleteStudentWithLessons(student.getId());
    }
}
```

Interviewer guidelines

Candidate has to know how to use spring data in any spring application which uses spring data. If an entity wanted to remove from database without removing the child attributes, then first relation between them should be removed. Without doing this, spring will throw DataIntegrityViolationException to keep database consistent. To achive that without changing existing code, we need to annotate deleteStudentWithLessons(Long id) method with @Transactional annotation. With this spring will update the database states of child entities. In this case it will update Lesson attributes to have student as null. Then student will be removed in the next line.

After this there is a inProgress check for existing lessons. it will throw SQLException which is a checked exception. @Transactional annotation in default only rolls back for unchecked exception such as RuntimeException. So all changes will be reflected to database.

Option A and E is right. More info can be found in https://docs.spring.io/spring-framework/docs/current/javadoc-api/org/springframework/transaction/annotation/Transactional.html

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [ ] It will try to delete the student then the DataIntegrityViolationException will be thrown.

- [x] ** It will delete the student successfully then an SQLException will be thrown with the message "Cannot delete all Lessons. There are lessons in progress state". **

- [ ] If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException and all database changes will be rolled back.

- [ ] It will throw a RuntimeException with the message "Student not found".

- [x] ** If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException. **

No comments.

5. Endpoint Security

Correct

Multiple Choice | Spring Boot | Medium

Question description

In Spring Boot Security, the option to enable secure endpoints with @PreAuthorize annotation is:

Interviewer guidelines

@EnableGlobalMethodSecurity by default doesn't enable this feature. To enable this prePostEnabled=true value must be set

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ Configuration class annotated with @EnableGlobalMethodSecurity
- ○ Configuration class annotated with @EnableGlobalMethodSecurity(securedEnabled = true)
- ● **Configuration class annotated with @EnableGlobalMethodSecurity(prePostEnabled = true)** ✓
- ○ None above

No comments.

6. Wind Farms

Correct

Coding | Implementation | Medium | Arrays | Algorithms | Problem Solving

Question description

An energy company needs to place a control center to manage a distributed wind farm. Your task is to determine the optimal location for the control center that minimizes the total connection cost.

For each wind farm *i*:

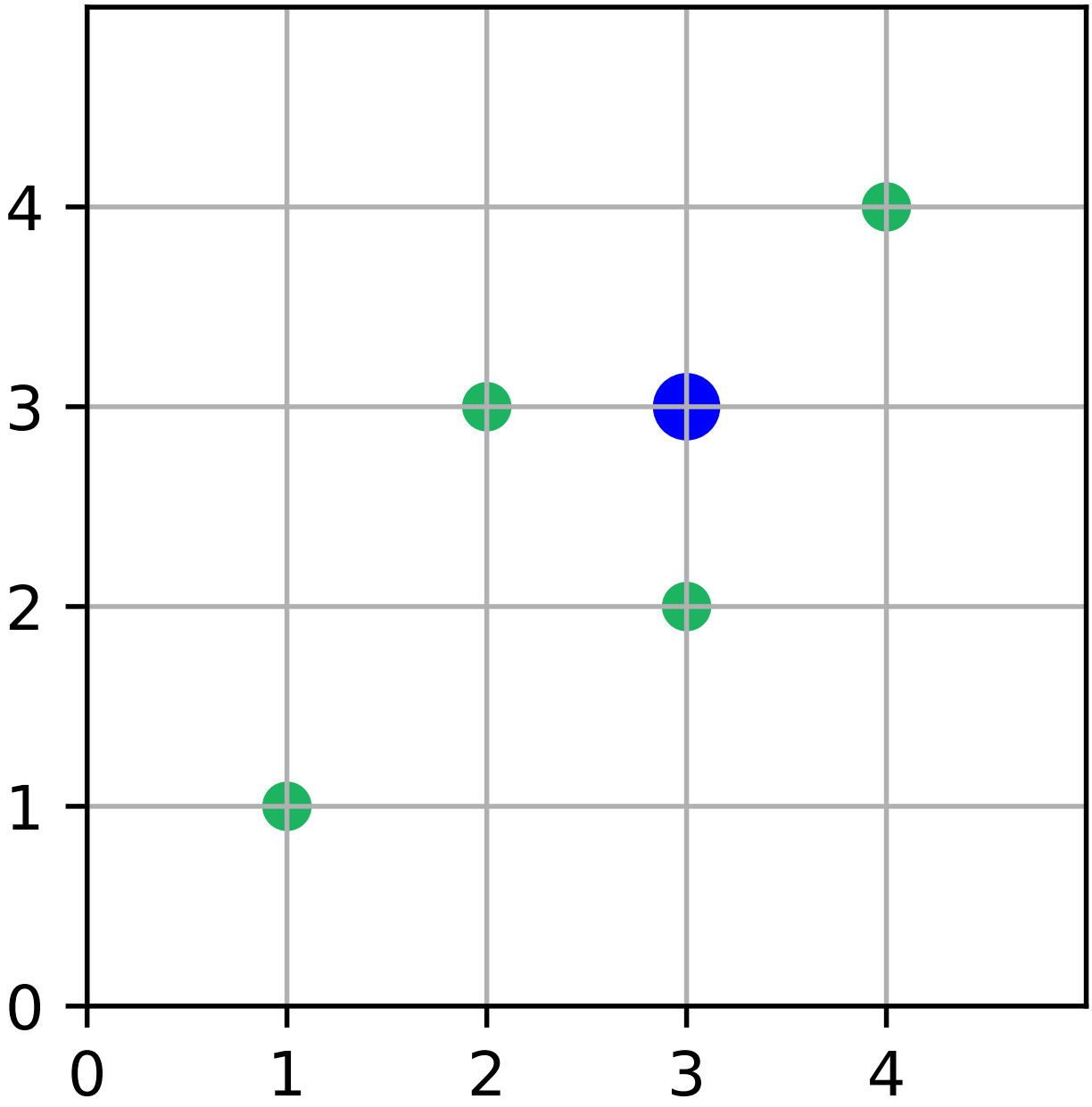
- The farm is located at coordinates *(x[i], y[i])*
- The control center is at coordinates *(cx, cy)*
- The distance is calculated as *|x[i] - cx| + |y[i] - cy|*
- The connection cost is *premium[i] * distance*

Example

x = [1, 3, 2, 4]

y = [1, 2, 3, 4]

premium = [1, 3, 2, 4]



The optimal location is at position [3, 3], resulting in:

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 3| = 4	1	4 * 1 = 4

1	(3, 2)	|3 - 3| + |2 - 3| = 1	3	1 * 3 = 3
2	(2, 3)	|2 - 3| + |3 - 3| = 1	2	1 * 2 = 2
3	(4, 4)	|4 - 3| + |4 - 3| = 2	4	2 * 4 = 8

The total connection cost is 17, which is the minimum possible.

Function Description

Complete the function *windFarms* in the editor with the following parameter(s):

int premium[n]: the distance multiplier for the cost to connect
int x[n]: *x* coordinates of the wind farms
int y[n]: *y* coordinates of the wind farms

Returns

int: the minimum cost to connect the wind farms

Constraints

- $1 \le n \le 10^3$
- $1 \le x[i], y[i] \le 10^4$
- $1 \le premium[i] \le 50$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer *n*, the size of the array *premium*.
The next *n* lines each contain an element, *premium[i],* where 0 ≤ i < n.
The next line contains an integer *n*, the size of *x[]*.
The next *n* lines each contain an element, *x[i]*.
The next line contains an integer *n*, the size of *y[]*.
The next *n* lines each contain an element, *y[i]*.

Note: *n* is the same value in each instance.

▼ SAMPLE CASE 0

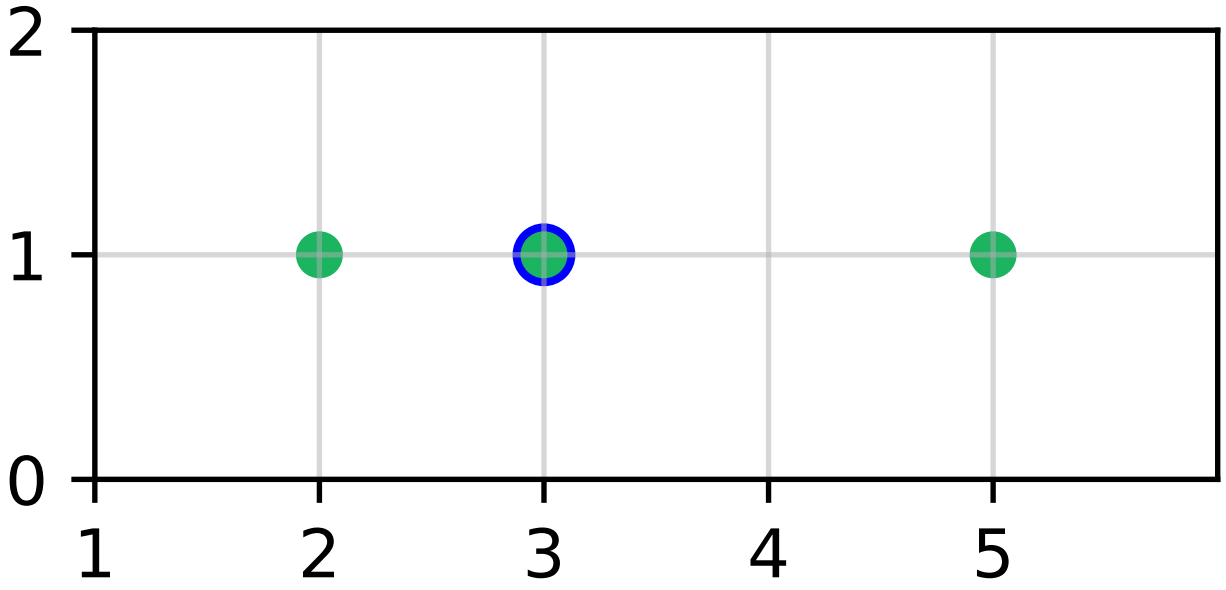
Sample Input 0

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
2    →  x = [2, 3, 5]
3
5
3    →  y[] size n = 3
1    →  y = [1, 1, 1]
1
1
```

Sample Output 0

```
3
```

Explanation 0



Locate the controller at farm 1.

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 1| = 1	1	1 * 1 = 1
1	(3, 1)	|3 - 3| + |1 - 1| = 0	1	0 * 1 = 0

2	(5, 1)	|5 - 3| + |1 - 1| = 2	1	2 * 1 = 2

▼ SAMPLE CASE 1

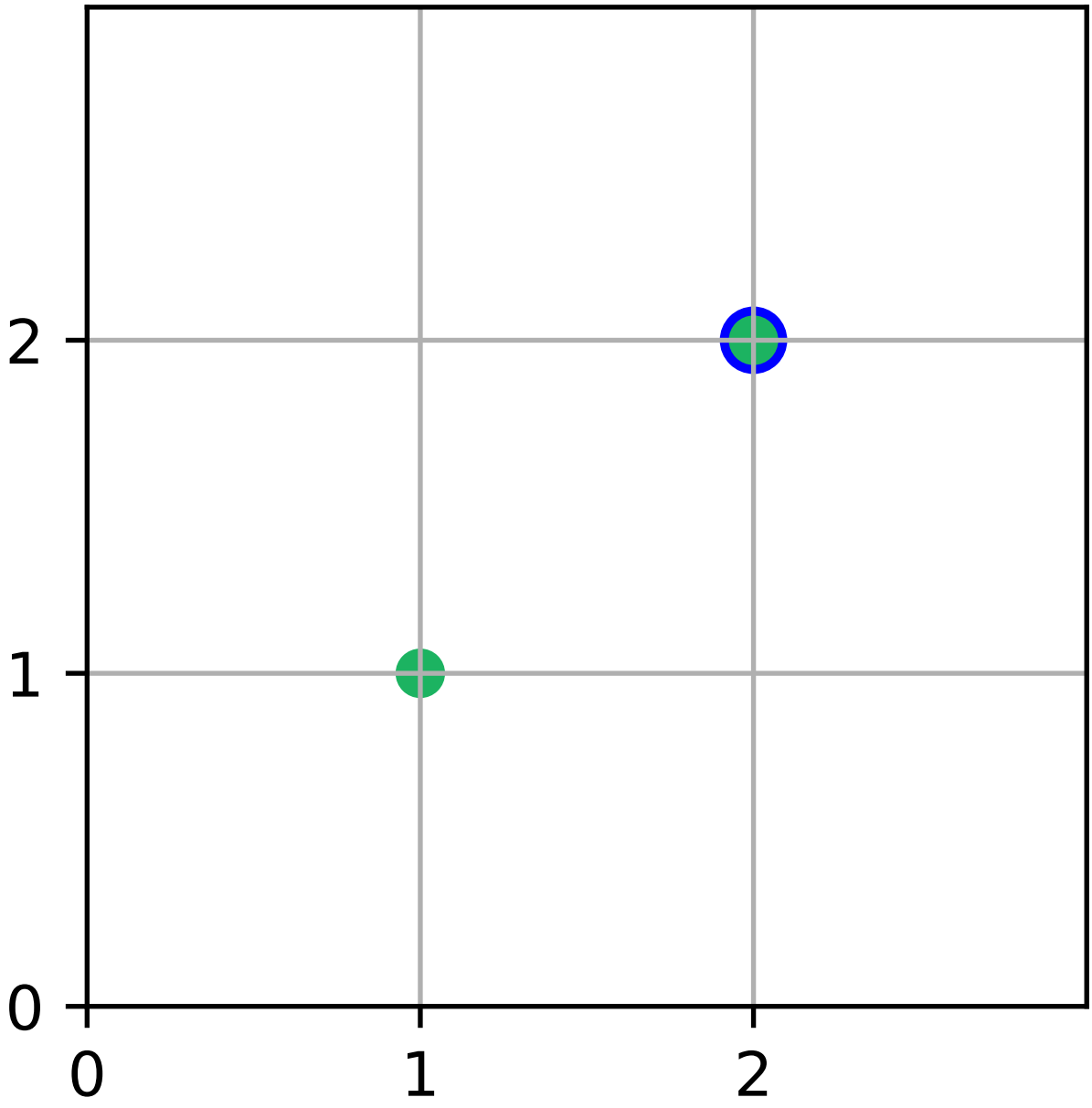
Sample Input 1

```
STDIN    Function
-----    --------
2    →  premium[] size n = 2
1    →  premium = [1, 10]
10
2    →  x[] size n = 2
1    →  x = [1, 2]
2
2    →  y[] size n = 2
1    →  y = [1, 2]
2
```

Sample Output 1

```
2
```

Explanation 1



The sum of the distances to a center at (1, 1), (1, 2), (2, 2), or (2, 1) is 2. Since the premium to connect farm 1 is 10, it is best to locate at position (2, 2) for a cost of 2 * 1 + 0 * 10 = 2.

▼ SAMPLE CASE 2

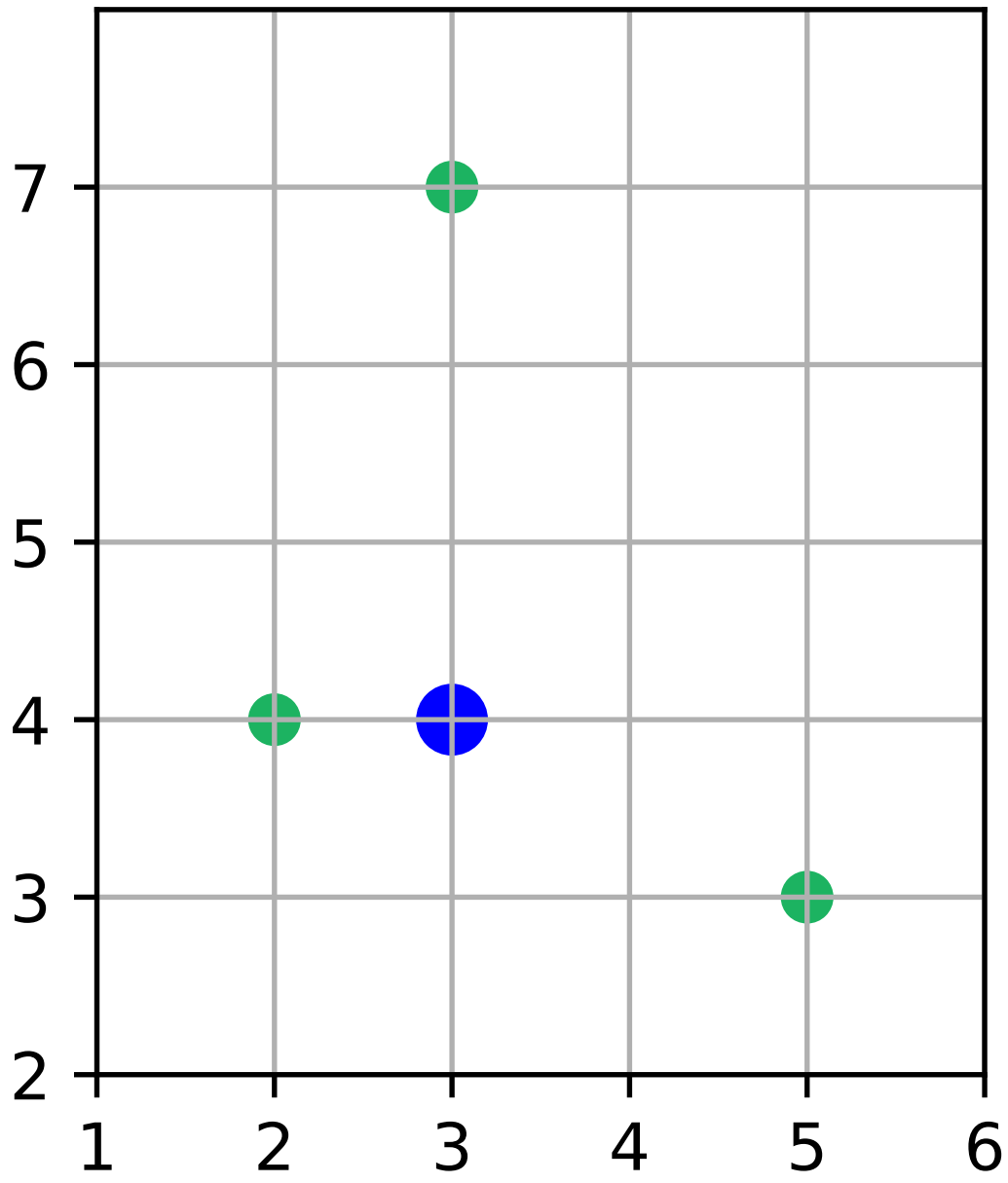
Sample Input 2

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
5    →  x = [5, 2, 3]
2
3
3    →  y[] size n = 3
3    →  y = [3, 4, 7]
4
7
```

Sample Output 2

```
7
```

Explanation 2



The optimal location is (3, 4) for a cost of 3 + 1 + 3 = 7.

Interviewer guidelines

▼ SOLUTION

Skills: Loops and counters, I/O

The cost can be calculated for x and y coordinates separately. The function examines each possible x and y coordinate for the control center separately (from 1 to 10000), computes the total cost of connecting all farms for each possible coordinate, and captures the minimum cost found. The ultimate minimum cost is the summation of these minimum costs for x and y coordinates separately.

Optimal Solution

```
def windFarms(premium, x, y):
    minCostX = 10**10
    n = len(x)

    for cx in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cx - x[i])

        minCostX = min(minCostX, cost)

    minCostY = 10**10
    for cy in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cy - y[i])

        minCostY = min(minCostY, cost)

    return minCostX + minCostY
```

▼ **COMPLEXITY ANALYSIS**

Time Complexity - O(n)
The universe of x and y values are tested against 10000 values.

Space Complexity - O(1)
Constant space is required regardless of n.

Candidate's Solution

Language used: **Java 21**

```java
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


import java.io.*;
import java.lang.reflect.Array;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;



class Result {

    /*
     * Complete the 'windFarms' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER_ARRAY premium
     *  2. INTEGER_ARRAY x
     *  3. INTEGER_ARRAY y
     */

    public static int windFarms(List<Integer> premium, List<Integer> x, List<Integer> y) {
    // Write your code here
    int n= premium.size();
```

```java

    int cx= getWeightedMedian(x, premium);
    int cy = getWeightedMedian(y, premium);

    long totalCost=0;

    for(int i=0; i<n;i++){
        long distance = Math.abs(x.get(i)- cx) + Math.abs(y.get(i) - cy);
        totalCost+= (long) premium.get(i) * distance;
    }
    return (int) totalCost;
    }

    public static int getWeightedMedian(List<Integer> coords, List<Integer> weights){
        int n = coords.size();

        List<int[]> list = new ArrayList<>();
        for(int i=0;i< n;i++){
            list.add(new int[]{coords.get(i), weights.get(i)});
        }
        list.sort(Comparator.comparingInt(a-> a[0]));

        long totalWeight= 0;
        for(int w:weights) totalWeight+=w;

        long half = (totalWeight+1)/2;
        long cumulative =0;

        for(int[] pair : list){
            cumulative+= pair[1];
            if(cumulative >= half){
                return pair[0];
            }
        }
        return list.get(n-1)[0];

    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
```

```
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.windFarms(premium, x, y);

```

```
        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
```

```
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.windFarms(premium, x, y);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.0512 sec	37.8 KB
TestCase 1	Easy	Sample	Success	1	0.0483 sec	37.8 KB
TestCase 2	Easy	Sample	Success	1	0.0557 sec	37.9 KB
TestCase 3	Easy	Sample	Success	2	0.0521 sec	38 KB
TestCase 4	Easy	Hidden	Success	2	0.0524 sec	37.7 KB
TestCase 5	Easy	Hidden	Success	2	0.058 sec	37.6 KB

TestCase 6	Medium	Sample	Success	6	0.0587 sec	38 KB
TestCase 7	Medium	Hidden	Success	6	0.0599 sec	38.2 KB
TestCase 8	Medium	Hidden	Success	6	0.0619 sec	38.6 KB
TestCase 9	Hard	Hidden	Success	12	0.0933 sec	43.5 KB
TestCase 10	Hard	Hidden	Success	12	0.0757 sec	43.4 KB
TestCase 11	Hard	Hidden	Success	12	0.0901 sec	43.4 KB
TestCase 12	Hard	Hidden	Success	12	0.0837 sec	44 KB

No comments.

7. Java: Optimized Sequential Video Watching

Correct

Coding Medium Greedy Java

Question description

A learner is given a sequential playlist of *n* videos to enhance their knowledge. Implement a function to calculate the minimum time required to complete a total of *m* viewings while adhering to the platform's constraints.

System Constraints

- Sequential Viewing:

  - The learner must watch videos in the order they appear in the playlist.
  - To watch a video at index *i*, the learner must first watch all videos from index 0 to *i-1* at least once.
- Initial Full Viewing:
  - To view any video at index *i* for the first time, the learner must spend *firstWatch[i] + repeatWatch[i]* minutes. This ensures a complete understanding of the video.
- Subsequent Viewings:
  - For subsequent viewings of a video at index *i*, the learner spends only *repeatWatch[i]* minutes.
  - At any time, they can rewatch any video they have already watched.
- Minimum Total Watch Count:
  - The learner must achieve at least *m* total viewings across all videos.
  - Videos can be rewatched as desired, and not all videos need to be watched equally.

Return the minimum time required to complete *m* viewings of videos while following these constraints.

Example

If *n* = 4, *m* = 4, *firstWatch* = [1, 5, 9, 11], and *repeatWatch* = [2, 7, 10, 11], the optimal strategy is as follows:

- The learner watches the 0th video for the first time, spending *firstWatch*[0] + *repeatWatch*[0] = 1 + 2 = 3 minutes.
- The learner then rewatches the 0th video 3 more times in, spending *repeatWatch*[0]*3 = 2 * 3 = 6 minutes.
- The learner has successfully completed 4 viewings. Thus, the total time required is 3 + 6 = 9 minutes, and 9 is returned.

Function Description

Complete the function *findMinMinutes* in the editor below.

Function Parameters

int m: the total number of video viewings to complete

int firstWatch[n]: the minutes required to watch each video the first time

int repeatWatch[n]: the minutes required to rewatch each video

Returns

long: the minimum minutes required for the learner to reach their target number of viewings

Constraints

- $1 \le n \le 10^5$
- $1 \le m \le 10^9$
- $1 \le firstWatch[i], repeatWatch[i] \le 10^9$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer, *m*.
The next line contains an integer, *n,* the size of *firstWatch*.
Each of the next *n* lines contains an integer, *firstWatch[i]*.
The next line contains an integer, *n,* the size of *repeatWatch*.
Each of the next *n* lines contains an integer, *repeatWatch[i]*.

▼ SAMPLE CASE 0

Sample Input 0

```
STDIN    FUNCTION
-----    --------
5        m = 5
3        firstWatch[] size n = 3
6        firstWatch = [6, 8, 9]
8
9
3        repeatWatch[] size n = 3
4        repeatWatch = [4, 2, 10]
2
10
```

Sample Output 0

```
26
```

Explanation

The goal is to complete 5 viewings. One optimal strategy is:

- The learner watches the 0th video for the first time, so the time required is *firstWatch*[0] + *repeatWatch*[0] = 6 + 4 = 10 minutes.
- The learner watches the 1st video for the first time, so the time required is *firstWatch*[1] + *repeatWatch*[1] = 8 + 2 = 10 minutes.
- The learner then rewatches the 1st video 3 times. So the time taken = *repeatWatch*[1]*3 = 2 * 3 = 6 minutes.

- The total time required to complete 5 video viewings is 10 + 10 + 6 = 26 minutes.

▼ SAMPLE CASE 1

Sample Input 1

```
STDIN     FUNCTION
-----     --------
6         m = 6
5         firstWatch[] size n = 5
2         firstWatch = [2, 3, 7, 8, 4]
3
7
8
4
5         repeatWatch[] size n = 5
3         repeatWatch = [3, 6, 8, 7, 6]
6
8
7
6
```

Sample Output 1

```
20
```

Explanation

The goal is to complete 6 viewings. One optimal strategy is:

- The learner watches the 0th video for the first time in *firstWatch*[0] + *repeatWatch[0]* = 2 + 3 = 5 minutes.
- The learner then rewatches the 0th video 5 times in *repeatWatch[0]**5 = 3 * 5 = 15 minutes.
- The total time required to complete 6 viewings is 5 + 15 = 20 minutes.

Interviewer guidelines

▼ SOLUTION

Description

The greedy solution suggests that, at any given index, if the time to watch the video is less than at any other index between i+1 and n, the minimum total time will be the sum of the watch time up to index i-1, plus the repeat time of the ith video multiplied by (m-i).

Solution

```
class Result {

    /*
     * Complete the 'findMinMinutes' function below.
     *
     * The function is expected to return a LONG_INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER m
     *  2. INTEGER_ARRAY firstWatch
     *  3. INTEGER_ARRAY repeatWatch
     */

    public static long findMinMinutes(int m, List<Integer> firstWatch, List<Integer> repeatWatch) {
        long minMinutes = Long.MAX_VALUE;
        long cur = 0;
        int n = firstWatch.size();

        for (int i = 0; i < Math.min(n,m); i++) {
            int a = firstWatch.get(i);
            int b = repeatWatch.get(i);

            minMinutes = Math.min(minMinutes, cur + ((long) (m - i) * b) + a);

            cur += a + b;
        }
        return minMinutes;
    }

}
```

▼ COMPLEXITY

Time Complexity - O(n)

The algorithm iterates through each number in *firstWatch* and *repeatWatch* exactly once, performing constant-time checks and arithmetic operations in each iteration. Thus, the time complexity is O(n), where *n* is the length of *firstWatch*.

Space Complexity - O(1)

Space usage is constant, only utilizing a fixed number of variables (*mn, cur,* etc.) that do not scale with the length of *firstWatch*. Therefore, the space complexity is O(1).

Candidate's Solution

Language used: **Java 21**

```java
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;



class Result {

    /*
     * Complete the 'findMinMinutes' function below.
     *
     * The function is expected to return a LONG_INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER m
     *  2. INTEGER_ARRAY firstWatch
     *  3. INTEGER_ARRAY repeatWatch
     */

    public static long findMinMinutes(int m, List<Integer> firstWatch, List<Integer> repeatWatch) {

        int n = firstWatch.size();
        long totalTime=0;
        long minTime=Long.MAX_VALUE;
        long minRepeat = Long.MAX_VALUE;

        int totalViews =0;

        for(int i=0;i<n;i++){
            totalTime += firstWatch.get(i) + repeatWatch.get(i);
            totalViews++;

            minRepeat = Math.min(minRepeat, repeatWatch.get(i));
            if(totalViews>=m){
                minTime = Math.min(minTime, totalTime);
            }
```

```
            else{
                long remaining = m- totalViews;
                long candidate = totalTime + remaining* minRepeat;
                minTime = Math.min(minTime, candidate);
            }
        }

        return minTime;

    // Write your code here

    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int m = Integer.parseInt(bufferedReader.readLine().trim());

        int firstWatchCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> firstWatch = IntStream.range(0, firstWatchCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int repeatWatchCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> repeatWatch = IntStream.range(0, repeatWatchCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
```

```
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        long result = Result.findMinMinutes(m, firstWatch, repeatWatch);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 2	Easy	Sample	Success	1	0.049 sec	37.8 KB
Testcase 0	Easy	Sample	Success	1	0.0489 sec	37.4 KB
Testcase 1	Easy	Sample	Success	1	0.0492 sec	37.3 KB
Testcase 3	Easy	Hidden	Success	4	0.05 sec	37.1 KB
Testcase 4	Easy	Hidden	Success	4	0.0595 sec	38.4 KB
Testcase 5	Easy	Hidden	Success	4	0.097 sec	45.6 KB
Testcase 6	Medium	Hidden	Success	5	0.2384 sec	71.4 KB

Testcase 7	Medium	Hidden	Success	5	0.6027 sec	167 KB
Testcase 8	Medium	Hidden	Success	5	0.5757 sec	147 KB
Testcase 9	Medium	Hidden	Success	5	0.564 sec	169 KB
Testcase 10	Medium	Hidden	Success	5	0.5384 sec	174 KB
Testcase 11	Medium	Hidden	Success	5	0.5342 sec	149 KB
Testcase 12	Hard	Hidden	Success	10	0.2374 sec	70.3 KB
Testcase 13	Hard	Hidden	Success	10	0.45 sec	116 KB
Testcase 14	Hard	Hidden	Success	10	0.0539 sec	37.8 KB

No comments.